

August 24

$$f(x) \rightarrow f'(x), \quad \frac{df}{dx}, \quad \frac{d}{dx}(f(x))$$

$$f(x) = 2 \Rightarrow f'(x) = 0$$

Power rule:

$$\frac{d}{dx}(x^{3/4}) = \frac{3}{4} x^{-1/4} \quad x^{3/4} = \sqrt[4]{x^3} \quad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}\left(\frac{1}{x^2}\right) = -2x^{-3}$$

$$\frac{d}{dx}\left(\sqrt[3]{x^2}\right) = \frac{2}{3} x^{-1/3} = \frac{2}{3\sqrt[3]{x}}$$

$$\frac{d}{dx}(e^{2x} - 1) = 2e^{2x}$$

$$f(x) = x \ln(x), \quad f'(x) = 1 + \ln(x)$$

$$\begin{aligned} \frac{d}{dx}\left(\frac{\sin(x)}{1 + \cos x}\right) &= \frac{\cos(x)(1 + \cos(x)) - (-\sin(x))\sin(x)}{(1 + \cos(x))^2} \\ &= \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{1 + \cos x}{(1 + \cos x)^2} \\ &= \frac{1}{1 + \cos x} \end{aligned}$$

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$$(f(g(x)))' = g'(x) \cdot f'(g(x))$$

$$\begin{aligned} (f(x) \cdot g(x))' &= f'(x)g(x) + f(x)g'(x) \\ &= f(x) \cdot g'(x) + g(x)f'(x) \end{aligned}$$

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - g'(x)f(x)}{(g(x))^2}$$

$$\frac{d}{dx} (5^{x^2}) = 5^{x^2} \ln 5 \cdot (2x)$$

$$\frac{d}{dx} (5^x) = 5^x \ln 5$$

$$5^{x^2} = e^{\ln 5^{x^2}} = e^{x^2 \ln 5}$$

Definitions:

Antiderivative : $f(x)$, $F(x)$ is an antiderivative of $f(x)$
 $\Rightarrow F'(x) = f(x)$

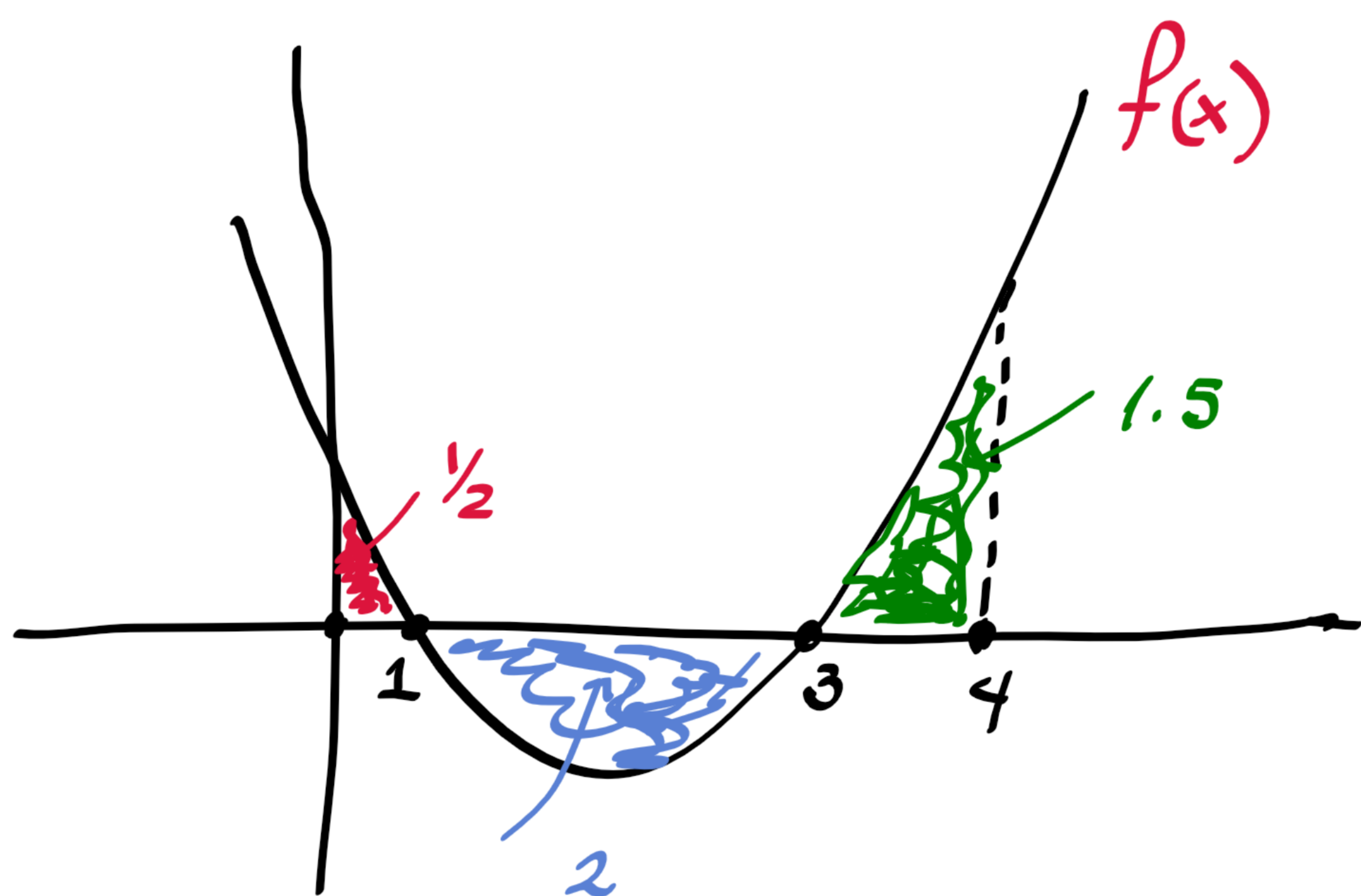
Indefinite integral: *family of antiderivatives of $\sin(x)$.*

$$\int \sin(x) dx = -\cos(x) + C$$

$$-\cos(x) + 1$$

$$-\cos(x) + \pi$$

Definite integral : $\int_a^b f(x) dx = \text{signed area.}$
 real numbers
 (limits of integration)



$$\int_0^1 f(x) dx = \frac{1}{2}$$

$$\int_1^3 f(x) dx = -2$$

$$\int_3^4 f(x) dx = 1.5$$

Some properties of definite integrals:

(Other properties can be found in the guided notes. We will write them when we need them).

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

c : between a and b .

Fundamental Theorem of Calculus (FTC-2)

$$\int_a^b f(x) dx = F(b) - F(a) \quad (\text{Part 2})$$

a, b : real numbers, $a < b$.

f is continuous on $[a, b]$.

F : one antiderivative of f .

~~$\int_{-1}^1 \frac{1}{x} dx$~~ FTC-2 cannot be applied because $\frac{1}{x}$ is not continuous on $[-1, 1]$

$\int_1^2 \frac{1}{x} dx$ ✓ (FTC-2 can be applied)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$n \neq -1$

~~$\int_0^{\pi/2} \tan(x) dx$~~ FTC-2 cannot be applied because $\tan(x)$ is discontinuous at $\pi/2$.

$\int_0^2 \sqrt{x} dx$ ✓ FTC-2 can be applied.

Substitution rule:

$$\int f(g(x)) g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C$$

← antiderivative of f

for indefinite integrals

$$\begin{cases} u = g(x) \\ \frac{du}{dx} = g'(x) \Rightarrow du = \underline{\underline{g'(x) dx}} \end{cases}$$

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du = F(u) \Big|_{g(a)}^{g(b)} = F(g(b)) - F(g(a))$$

for definite integrals

$$\begin{cases} u = g(x) \\ du = g'(x) dx \end{cases}$$